Tutorial 2

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1. $P(b) = 1 - P(a) - P(c) - P(d) - P(e)$

$= 0,15$

$P(A) = P(c) + P(d)$

$= 0,5$

$P(A') = 1 - P(A) = 0,5$

2.

Let A be the event that we roll 2 dices continously and the red dice has the strictly greater score than the blue one.

$P(A) = \frac{15}{36} \approx 0,42$

Explanation:

When we roll 2 dices, the size of the sample

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space is 36 which include:

- 6 situations in which red dice's point is equal blue die's point such as (1;1); (2;2), (3;3);...
- 15 in which red's point > blue's point and 15 in which blue's point > red's point

So $P(A) = \frac{n(A)}{n(\Omega)} = \frac{15}{36}$

⇒ The probability would be less than 0,5.

+) The complement of the event would be:

$P(A') = 1 - \frac{15}{36} = \frac{7}{12}$

3.

Let A be the event of drawing a card of spade and B be the event of drawing a card club.

Then $P(A \cup B) = P(A) + P(B)$

$= \frac{13}{52} + \frac{13}{52} = \frac{26}{52} = \frac{1}{2}$

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4) Let A be the event that 2 heads will be obtained in succession when tossing 3 coins:

$\Rightarrow A = \{(H;H;T);(T;H;H);(H;H;H)\}$

$\Rightarrow P(A) = \frac{3}{8}$

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5. Take ~~set~~ event A into consideration. Set A has one element which is also included in the sample space, so $P(A) \neq 0$. Therefore, $A \neq \emptyset$. (because $P(\emptyset) = 0$)

6.

a) $P(B) = 0{,}08 + 0{,}13 + 0{,}06 + 0{,}01 + 0{,}02 + 0{,}05 + 0{,}11$
$= 0{,}46$

b) $P(B \cap C) = 0{,}02 + 0{,}05 + 0{,}11$
$= 0{,}18$

c) $P(A \cup C) = P(A) + P(C) - P(A \cap C)$
$= 0{,}61$

d) $P(A \cap B \cap C) = 0{,}02 + 0{,}05 = 0{,}07$

e) $P(A \cup B \cup C) = 1 - P((A \cup B \cup C)')$
$= 1 - (0{,}04 + 0{,}05 + 0{,}03)$
$= 0{,}88$

f) $P(A' \cap B) = P(B) - P(A \cap B)$
$= 0{,}46 - (0{,}01 + 0{,}02 + 0{,}05)$
$= 0{,}38$

g) $P(B' \cup C) = P(B') + P(B \cap C)$
$= 1 - 0{,}46 + 0{,}02 + 0{,}05 + 0{,}11$
$= 0{,}72$

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h) $P(A \cup (B \cap C)) = P(B \cap C) + P(A) - P(A \cap B \cap C)$

$= 0{,}18 + 0{,}32 - 0{,}07$

$= 0{,}43$

i) $P((A \cup B) \cap C) = P(C \cap A) + P(B \cap C)$

$- P(A \cap B \cap C)$

$= 0{,}19 + 0{,}18 - 0{,}07$

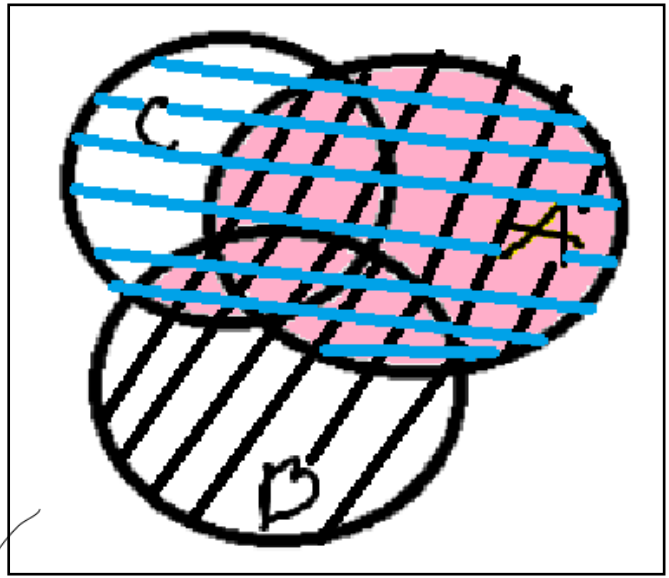
$= 0{,}3$

j) $P((A' \cup C)') = 1 - P(A') - P(A \cap C)$

$= 1 - 0{,}68 - 0{,}08 - 0{,}04$

$- 0{,}02 - 0{,}05$

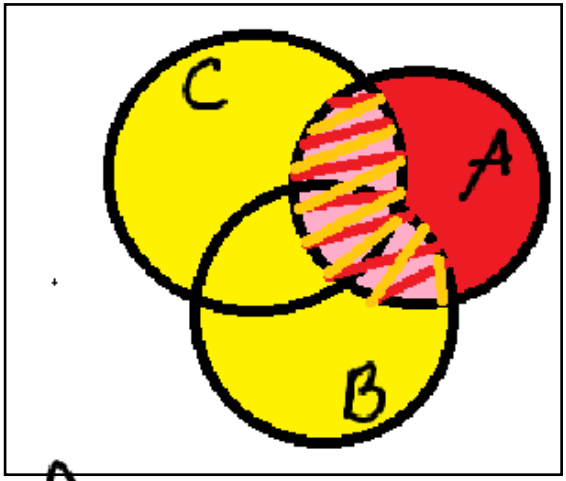
$= 0{,}13$

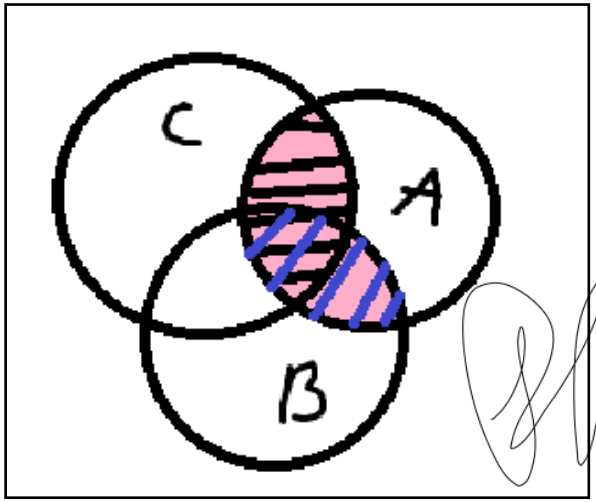


7a

- A
- $B \cap C$
- $A \cup (B \cap C)$

- $A \cup B$
- $A \cup C$
- $(A \cup B) \cap (A \cup C)$



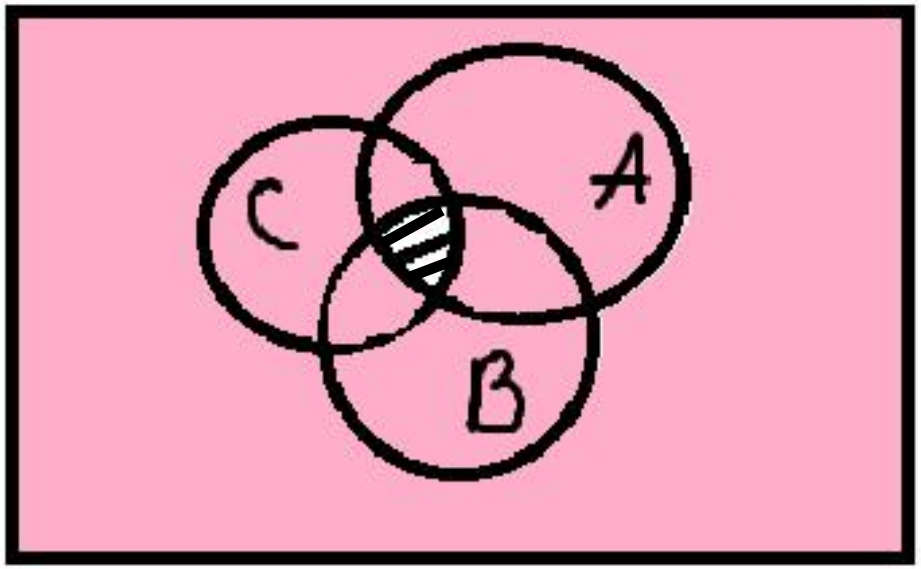


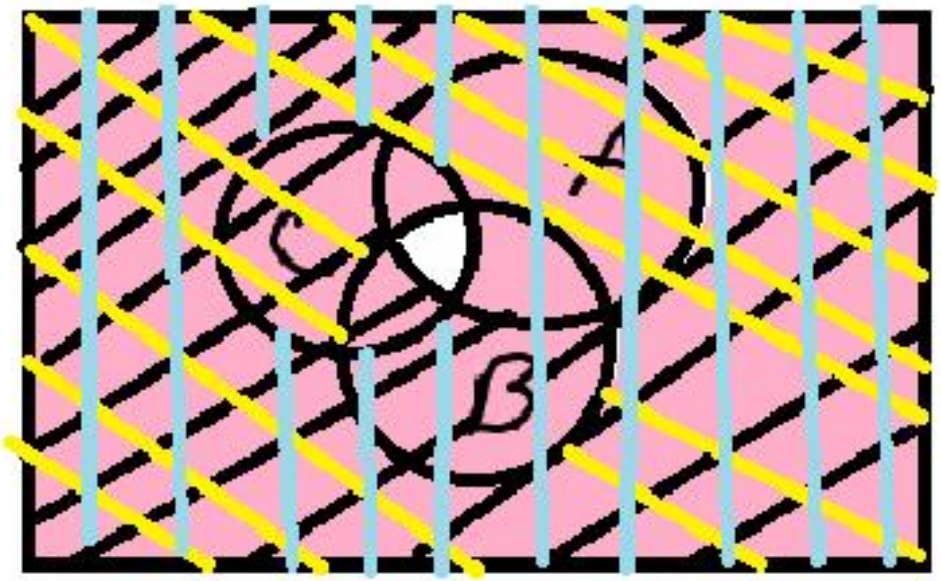
7b

- A
- $B \cup C$
- $A \cap (B \cup C)$

- $A \cap C$
- $A \cap B$
- $(A \cap C) \cup (A \cap B)$

7c





$A \cap B \cap C$

$(A \cap B \cap C)'$

A'

C'

B'

$A' \cup B' \cup C'$

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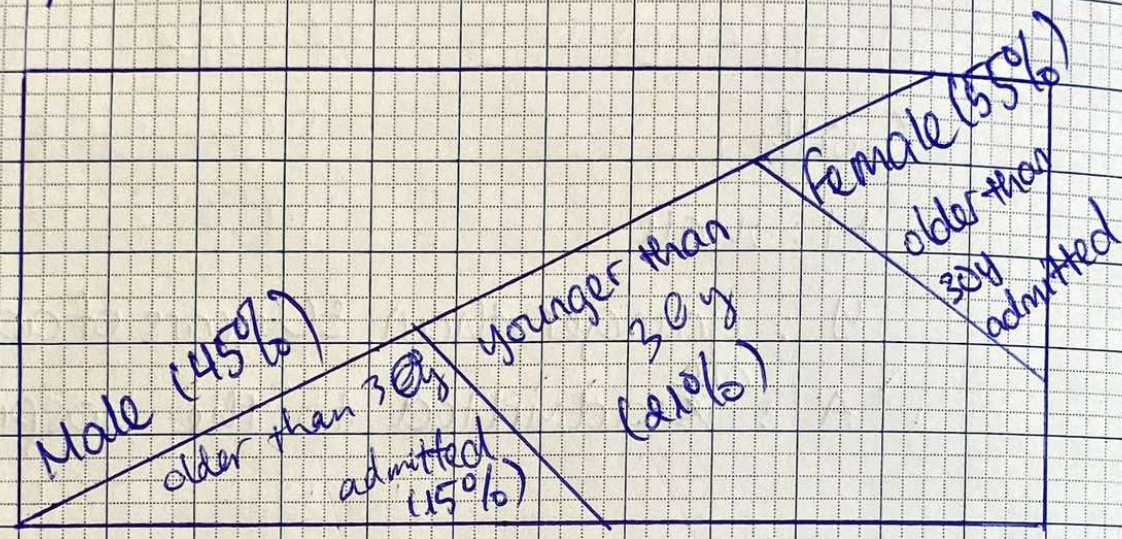
8.

According to the given statistics, we have a diagram:



The proportion of female patients who are older than 30 years of age and admitted to the hospital would be:

$$1 - 45\% - 21\% - 15\% = 19\%$$

Another way is

$$P(M' \cap y) + P(M' \cap y') = P(M') = 1 - P(M) = 0{,}55$$

$$\Rightarrow P(M' \cap y') = 0{,}55 - P(M' \cap y) = 0{,}55 - 0{,}21 = 0{,}34$$

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$P(M' \cap Y') = P(M' \cap Y' \cap A) + P(M' \cap Y' \cap A')$

$\Rightarrow 0{,}34 = 0{,}15 + P(M' \cap Y' \cap A')$

$\Rightarrow P(M' \cap Y' \cap A') = 0{,}19$

* with :

M : male

Y : younger than 30 years of age

A : are admitted to the hospital